Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Application of Nanocrystals and Nano-hybrids having Innovating **Up**conversion properties for **Security**-critical Documents and Components (**UpSecurity**) 보안 문서 및 핵심 부품을 위한 혁신적 상향 변환 광학 특성 나노결정의 응용 기술			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication (X), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		**Chemicals**	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Service and Application Security	Fine Chemicals	Nano-Application Technologies

Concept of Proposed Technology	**Core concept:** This project innovates by engineering multiple independent security features into a single platform for anti-counterfeiting application which is centered on **upconverting (UC) luminescent/phosphor nanomaterials** that absorb invisible near-infrared (NIR) light and re-emit it as customized visible light and/or NIR (Auzel, 2004). These materials can be engineered to produce unique optical "fingerprints," including multi-level emission modes that change color and unique lifetime signatures that create a time-based security code from the afterglow duration (You, 2024). Furthermore, a combination of NIR dyes will be integrated with UC nanophosphor as **nano-hybrids** that serve as antennas to enhance the upconversion luminescence. **Main functions:** The goal is to establish a technology platform that sets a new, higher standard in anti-counterfeiting and key innovations are, ▪ Novel UC nanophosphors and UC nanophosphor-NIR dyes as nano-hybrids ▪ Security inks consisting of UC nanophosphors and nano-hybrids ▪ Multi-dimensional optical finger printing ▪ Consumer friendly authentication ▪ Full integration from Materials-to-Market **Potential uses and application fields** ▪ Security documents, ID cards and currency ▪ High-value product labels and packaging; luxury goods and high-end consumer products ▪ Pharmaceuticals and medical devices ▪ Authentication markings for electronic, sensitive industrial and semiconductor components
Rationale for Joint R&D Project	**Global market requirement:** The global counterfeit goods trade, valued at up to $990 billion (United Nations report-UNODC; OECD 2025), causes significant socio-economic and health problems, which are amplified by the rise of e-commerce. To address this global market requirement, this project aims to create advanced optical security features that are hard to copy but easy to verify, moving beyond the limitations of traditional methods. **Dominance of South Korean industry and strategic gap:** While South Korea is a global leader in phosphor-related technologies such as quantum dots, display materials, and lighting, there's a strategic need to bolster its expertise in upconversion nanophosphors and their use in anti-counterfeiting applications. **Collaborative requirements:** This project requires a synergy of expertise in material development, processing, application testing, and market analysis. Fraunhofer IAP will provide its leading capabilities in upconversion nanophosphor engineering and micro-patterning. Through collaboration with Korean SMEs, these core technologies will be adapted for industrial-scale production. This partnership aims to establish a new security-material platform and strengthening the Korean industry's position in anti-counterfeiting. This collaborative effort will bridge the gap from lab to market, accelerate commercialization, reduce risks, and create a continuous innovation feedback loop through strategic market access and supply chain integration.

Key Words	▪ Security-critical documents and components for information and communication ▪ Anti-counterfeiting for information security ▪ Upconversion nanocrystals/nanophosphors ▪ NIR dyes ▪ Novel UC-Dye nano-hybrids ▪ Compatible with mobile devices ▪ Track and trace

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6/7)
R&D Objectives	▪ Engineered UC nanocrystals: specially designed rare earth doped UC nanocrystals/nanophosphors ▪ NIR dyes, high performance printable security inks, nano-hybrids and use of specific printing technology ▪ Evaluation of optical-properties and anti-counterfeiting security level: customized spectral fingerprints ▪ Scale-up of material production and high-throughput printing ▪ Consumer level smartphone verification: Portable simple device compatible with smart phones			
R&D Target Specifications	The target performance specifications for each key technological component are defined by the following quantitative metrics: ▪ UC nanophosphor performance: optical properties of UC nanocrystals need to match demands of customers/application with regards to detector system (absorption and emission wavelengths, emission intensity, fluorescence lifetime): UC efficiency > 5% quantum yield for NIR-to-visible conversion under a standard excitation power density (< 10 W/cm²). ▪ NIR dye performance: The selected NIR dyes must act as efficient antennas for the upconverting nanocrystals. This requires strong absorption at NIR wavelength and an energy transfer efficiency to the UC nanocrystals of >30%. ▪ Nano and nano-hybrids consisting of security ink formulation performance: stable ink formulation with; a shelf life of > 6 months at room temperature, showing no visible nanoparticle aggregation or phase separation; optimized viscosity for the printability; adhesion and substrate compatibility: ▪ Printed security feature performance: by targeting printing resolution at least 600 dots per inch (DPI) with invisible features and durability of printed features must withstand >100 cycles of abrasion tests and demonstrate resistance to common solvents. ▪ Authentication system performance: consumer level verification with smartphone within a fraction of seconds <5 seconds and ambient light tolerance. In professional level verification the authentication accuracy should be >99,9%.			
R&D Plan	**Research approach:** Phase 1: Scalable material synthesis and optimization Phase 2: Formulation of security inks and printing Phase 3: Application development and scaling Phase 4: Performance evaluation and validation; further market applicability **Key tasks, methodologies and expected milestones:** WP1: Development of UC nanophosphors (project month 1 – 36; Lead partner: Fraunhofer IAP): Goal is to establish a robust, scalable wet-chemical synthesis for manufacturing high quality UC nanocrystals. This WP includes work on synthesizing high-efficient engineered UC nanocrystals for multilevel excitation by co-doping with different rare earth ions and conduct comprehensive characterization using Transmission Electron Microscopy (TEM) for morphology, Dynamic Light Scattering (DLS) for size distribution, X-ray Diffraction (XRD) for crystal structure, and photoluminescence spectroscopy for optical properties. Expected milestones: UC nanocrystals demonstrating >3% (M1; 12th Month) and >5% (M2; 24th Month) quantum efficiency.			

WP2: NIR dyes development and nano and non-hybrid consisting security ink formulation and process optimization (project month 7– 36; Lead partner: Fraunhofer IAP with a Koren Partner): Goal is to develop IR dyes and establish novel ink formulations containing UC nanocrystals only and also in combination with IR dyes as hybrid material systems for use on different surfaces (paper, plastics, metal) and in different matrixes (esp. polymer). WP includes development of chemical surface functionalization methodologies to ensure stable UC nanocrystal and nano-hybrid dispersions in ink composition; formulate stable inkjet, EHD jet, gravure, slot-die and bar coating inks; comparison of printing technologies with regards to printing quality, testing of printed security marking for robustness, long term stability and uniqueness.

Expected milestones: Successful design and synthesis of NIR dyes meeting target spectroscopic properties (M3; 18th Month); integration of NIR dyes with UC nanocrystals, demonstrating effective sensitization with high energy transfer efficiency (> 30%) (M4; 30th Month). Stable ink formulation with appropriate rheological properties with shelf-life of > 3 months (M5; 18th Month); finalized and validated ink formulation meeting all performance targets (M6; 30th Month).

WP3: High-resolution feature printing (patterning) and integration (project month 1– 36; Lead partner: Fraunhofer IAP): To use the developed inks to print high-complexity, durable and reliable security features and integrate them into relevant formats. This includes establishing of printing and coating processes (e.g., inkjet, EHD-jet, gravure, slot-die, or bar coating) for UC composite and nano-hybrid layers and development of micro-patterning technologies for security and functional features.

Expected milestones: Successful printing of a 600 DPI invisible (to human eye) pattern that is readable by a lab-based authentication system (M7; 20th Month) and prototype security label produced on a flexible film that passes all specified durability and reliability tests (M8; 30th Month).

WP4: Industrialization strategy and market alignment (project month 1– 36; Lead partner: Korean SME): To ensure the R&D outcomes are commercially viable, manufacturable at scale, and continuously aligned with target market needs throughout the entire project lifecycle, thereby maximizing the potential for successful technology transfer and market adoption. Key tasks are analysis of market requirement and cost performance, intellectual property and regulatory matters, pilot line technology transfer and business case and go to market plan.

Expected milestones: commercial viability framework (M9: 15th Month) and final commercialization plan (M10:36th Month)

WP5: Authentication system development, function verification and validation (project month 12– 36; Lead Partner: Korean SME, with other partners): To develop, integrate, and rigorously validate the dual-mode authentication system (both consumer-level and professional-level) to ensure it meets all specified performance, reliability, and security targets, culminating in a fully verified and documented technology platform. This includes designing and building the hardware for the professional reader prototype- simple portable smart readable and conducting large-batch testing (>500 samples) to quantify authentication accuracy (false positive/negative rates), speed, and reliability under varied ambient conditions.

Expected milestones: Integrated system prototype ready for validation (M11; 30th Month) and final system validation (M12; 36th Month).

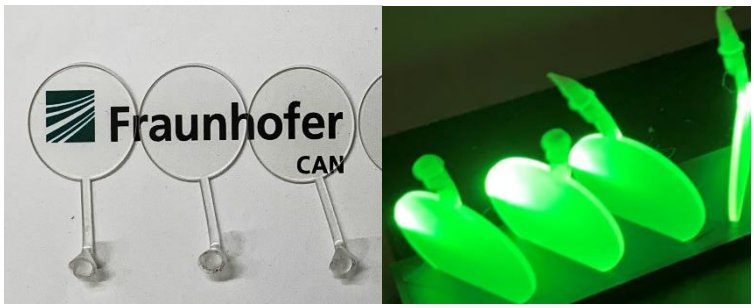
WP6: Further concepts for increasing application potential (project month 24– 36; All partners): To identify and perform proof-of-concept exploration for at least two new technological concepts that expand the functionality and market applicability of the project.

Expected milestones: A technology roadmap report outlining 3–5-year R&D pathways for future development (M13; 36th Month)

The timeline required is shown in Gannt Chart, see below

WP	Project Months	Lead Partners	1 year	2 year	3 year
			01 02 03 04 05 06 07 08 09 10 11 12	01 02 03 04 05 06 07 08 09 10 11 12	01 02 03 04 05 06 07 08 09 10 11 12
WP1	Development of UC nanophosphors	IAP	M1	M2	
WP2	NIR dyes, nano & nano-hybrid ink	IAP/Korean Inst.		M3&M5	M4&M6
WP3	High-resolution feature printing	IAP		M7	M8
WP4	Industrialization and market alignment	Korean SME		M9	M10
WP5	Authentication system &validation	Korean SME			M11 M12
WP6	Further concepts	All			M13

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	▪ Scalable manufacturing route for high quality UC nanocrystals (optimization regarding precursor chemicals, synthesis temperature, yield, etc.) ▪ Characterization of optical properties (spectra of excitation, emission, etc.) ▪ Formulation of UC security inks (binder, solvent, dispersant, matrix optimization) see below: UC nanocrystals incorporated polymer matrix; left) without IR source; right) with IR source (TRL3/4)  ▪ Development of high-resolution printing processes (including inkjet and EHD-jet) and optimization of printing process for security features ▪ Providing support regarding building demonstrator applications
Expected R&R of Korean SME Partner	▪ Design and development of IR dyes and nano-hybrid ink formulation (binder, solvent, dispersant, matrix optimization) ▪ Formulation design of UCNP security inks (binder, solvent, dispersant, matrix optimization) ▪ Scale-up and mass-production process development for UCNPs and inks with ensured batch-to-batch uniformity ▪ Demonstration of security devices or documents with advisory input from end users

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical Outcomes • New class of engineered nanomaterials: UC materials and printable / coatable composite layers with unique optical fingerprints. • Industrial grade printable security inks • High-resolution security patterns and films suitable for NIR-based authentication. • Demonstration of mobile-readable anti-counterfeiting technology. • Creation of new intellectual property either with (i) joint ownership of jointly developed results and (ii) an option for the industrial partner to negotiate an exclusive, royalty-bearing license Industrial and Commercial Impact • Establishment of a new high value product line security-material platform for a Korean SME. • Creation of prototype products for security documents, labels, and electronic components. • Foundation for future standardization and global market entry in advanced authentication materials. • Significant reduction in economic losses, which is estimated to be a multi hundred-billion-dollar global problem • Creation of new revenue and service models Broader Strategic and Socio-Economic Impact • Strengthening of Korea–Germany collaboration in advanced functional materials and security technologies, which will create a model for future collaborations that bridge cutting edge research with industrial scale production. • Contribution to high-value material and smart manufacturing industries through innovative photonic security solutions. • Expansion of international collaboration and market access, particularly toward the European security and authentication materials market. • By protecting legitimate industries from the economic damage of counterfeiting, securing the existing jobs and will also create new high skilled employment opportunities.

6. References

Auzel, 2004: F. Auzel, Upconversion and Anti-Stokes Processes with f and d Ions in Solids, Chem Rev. 2004, 104, 139

OECD 2025: Mapping Global Trade in Fakes 2025: Global Trends and Enforcement Challenges, Illicit Trade, OECD Publishing, Paris, https://doi.org/10.1787/94d3b29f-en.

You 2024: W. You, C. Zhang, R. Yu, X. Zhang, J. Li, M. Li, Z. Xu, P. Fan, G. Pan, Y. Mao, Excitation power-dependent multicolor upconversion in $NaLnF_4:Er^{3+}$ under 1532 nm irradiation for anti-counterfeiting application. Optics Letters, 2024, 11, 2978.

UNODC: Counterfeit_focussheet_EN_HIRES.pdf

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization